



Presentation Title

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A data structure is a way of organizing and storing data so it can be accessed and modified easily.

Data Structures and String Formatting:

1. Lists

A **list** is an ordered and mutable collection of items. It can store elements of different data types and allows duplicate values.

Syntax: `list_name = [item1, item2, item3]`

Example:

```
numbers = [10, 20, 30, 40]
print(numbers)
```

Output:

```
[10, 20, 30, 40]
```

2. Tuples

A **tuple** is an ordered and immutable collection of elements.

Syntax: `tuple_name = (item1, item2, item3)`

Example:

```
colors = ("red", "green", "blue")
print(colors)
```

Output:

```
('red', 'green', 'blue')
```

3. List Comprehension

A concise way to create lists.

Syntax: `[new_expression for item in iterable if condition]`

Example: `squares = [x*x for x in range(5)]
print(squares)`

Output:
`[0, 1, 4, 9, 16]`

4. Dictionaries

A dictionary stores data as **key-value pairs**.

Syntax: `dict_name = {
 key:value
}`

Example: `student = {"name":"alex",
 "age":22,
 "course":"Python"}
print(student)`

Output:
`{'name': 'alex', 'age': 22, 'course': 'Python'}`

5. Sets

A set is an unordered collection of unique elements.

Syntax:

```
set_name = {element1 , element2 , element3 }
```

Example: `numbers = { 1 , 2 , 3 , 4 , 4 , 5 , 5 , 6 , 3 , 2 }`
 `print(numbers)`

Output:

```
{1, 2, 3, 4, 5, 6}
```

Note: `my-dict =    #To create empty dictionary`
 `my_set = {}    #To create a empty dictionary in place of set`
 `my_set = set() #use to create empty set`

6. String Manipulation

A string is a sequence of characters enclosed in quotes.

upper()

```
print(name.upper())            #use to convert string having string_name name in upper case
```

lower()

`print(name.lower())` #use to convert string having string_name name in lower case

strip()

`text = " Hello "` #use to removes leading and trailing characters(such as
`print(text.strip())` spaces , tabs , and newline)from both ends of a string

7. f-string :

f-Strings provide an easy way to format strings and were introduced in Python 3.6.

Syntax: `f"text {variable}"`

Example :

```
a = 10
b = 20
print(f"Sum = {a+b}")
```

Output :

Sum = 30